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HybridMamba++: A Multi-Scale Series-Decomposed State-Space Model for High-Precision Bearing Anomaly Detection

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bullet.png **ABSTRACT** Prognostics and Health Management (PHM) for rolling element bearings is pivotal for the reliability of modern industrial assets. While state-of-the-art Channel-Independent (CI) architectures utilizing Mamba State-Space Models (SSMs) have shown promise in time-series forecasting, their application in bearing PHM remains hindered by high-frequency vibration transients and the non-linear nature of mechanical degradation. This paper proposes **HybridMamba++**, a novel architecture that integrates series decomposition with multi-scale dual-kernel convolutional patching. Unlike vanilla Mamba models, HybridMamba++ explicitly decouples the vibration signal into trend and seasonal components, processing them through independent paths to capture both long-term degradation trajectories and short-term impulse faults. To ensure statistical robustness, we employ the Peak-Over-Threshold (POT) method based on Extreme Value Theory (EVT) for automated anomaly threshold calibration. Benchmarked against LSTM, TCN, ModernTCN, and Transformers under a strict 250k parameter budget, HybridMamba++ achieves a superior F1-Score of 0.9859 and an AUPRC of 0.9944 on the PRONOSTIA (FEMTO-ST) dataset. Furthermore, it demonstrates an ultra-low inference latency of 0.0673 ms/sample, underscoring its suitability for real-time edge deployment in Industrial IoT (IIoT) environments.

bullet.png **INDEX TERMS** Anomaly Detection, Bearing Prognostics, Channel-Independent, Deep Learning, FEMTO-ST, Health Index, Hybrid State-Space Models, Mamba, PHM, Remaining Useful Life (RUL), Rolling Element Bearings.

I. INTRODUCTION

ROLLING element bearings serve as the "heart" of rotating machinery, yet they remain one of the most vulnerable components, accounting for approximately 45% to 70% of all electrical machine failures [1]. In the era of Industry 4.0, Prognostics and Health Management (PHM) has transitioned from reactive maintenance to proactive anomaly detection, leveraging high-frequency vibration data to identify incipient structural degradations before they escalate into catastrophic system failures.

Traditional deep learning approaches, such as Long Short-Term Memory (LSTM) and Temporal Convolutional Networks (TCN), have laid the foundation for automated feature extraction. However, these models often struggle with the "curse of long-range dependency"—where the signal length required to capture complete degradation cycles exceeds the effective receptive field of the network. While Transformers addressed this through self-attention, their quadratic computational complexity $O(L^2)$ poses a significant bottleneck for real-time edge monitoring where resources are constrained.

The recent introduction of the Mamba architecture [11], a selective State-Space Model (SSM), has redefined the efficiency-accuracy trade-off by achieving linear complexity $O(L)$. However, existing Mamba-based PHM models, such as FEMamba [13] and TFG-Mamba [14], primarily focus on direct Remaining Useful Life (RUL) regression. These approaches often overlook two critical physical characteristics of bearing vibration: (i) the non-stationary nature of signal components (Trend vs. Seasonal) and (ii) the multi-scale nature of fault signatures, which range from broad energy shifts to micro-second impulse transients.

To address these gaps, this paper introduces **HybridMamba++**, a Channel-Independent (CI) architecture specifically engineered for bearing PHM. The core contributions are:

1) **Series-Decomposed Hybrid Architecture**: We implement a decomposition layer that decouples the input signal into Trend and Seasonal components, allowing the model to learn stable degradation trajectories and volatile fault impulses separately.
2) **Multi-Scale Dual-Kernel Patching**: A novel front-end that captures multi-frequency transients using asymmetric convolutional kernels, significantly improving the sensitivity to early-stage faults.
3) **EVT-Based Threshold Calibration**: Integration of the Peak-Over-Threshold (POT) method for automated, statistically-grounded anomaly detection, achieving a near-zero False Alarm Rate (FAR) of 2.03%.
4) **Rigorous Hardware-Aware Benchmarking**: A comparative study under a strict 250k parameter budget, demonstrating that HybridMamba++ outperforms Transformers and ModernTCN in both precision (AUPRC: 0.9944) and inference speed (0.0673 ms/sample).

The remainder of this paper is organized as follows. Section II reviews foundational work in bearing PHM and Health Index modeling. Section III surveys related deep learning and Mamba-based approaches. Section IV describes the proposed CI-Mamba architecture and the experimental setup. Conclusions and future directions are discussed in Section VI.

II. LITERATURE REVIEW

The field of Prognostics and Health Management (PHM) for industrial components, particularly rolling element bearings, has undergone a significant transformation over the past decade. Rotating machinery is fundamental to modern industrial infrastructure, and bearing failure remains a primary cause of unplanned system downtime [1]. Traditionally, bearing health monitoring relied on physical modeling and statistical signal processing. Time-domain features, such as Root Mean Square (RMS) and Kurtosis, have served as foundational health indicators due to their sensitivity to early-stage impulse faults and late-stage energy increases [2].

A critical milestone in this domain was the introduction of the PRONOSTIA platform (FEMTO-ST dataset) [3], which provided standardized vibration data under varying operating conditions. Recent literature emphasizes the construction of a robust Health Index (HI) as a precursor to RUL prediction. While linear HI models were common, research has shown that mechanical degradation often follows an exponential trajectory, particularly in the final stages of failure [4]. This led to the adoption of exponential Health Indices, defined as:

$$HI(t) = \frac{e^{\lambda} - e^{\lambda \cdot \frac{t}{T}}}{e^{\lambda} - 1} \tag{1}$$

where λ represents the degradation rate and T is the total lifetime. Such formulations allow models to prioritize the capturing of accelerated degradation patterns near end-of-life, which are crucial for triggering timely maintenance alerts.

In 2025, the research landscape shifted toward **Mamba-based diagnostics**. Feng et al. [13] demonstrated that SSMs could capture long-term degradation stages more effectively than LSTMs through global regularization. However, the computational overhead of multivariate scanning in models like MambaTS often limits their deployment on low-power industrial controllers. Our work builds upon the **Channel-Independent (CI) paradigm** introduced in PatchTST [10], which suggests that processing sensors independently can prevent channel-mixing noise and improve forecasting stability in PHM tasks.

An anomaly is declared when the model-predicted HI falls below a calibrated threshold θ, defined as:

$$\hat{y}_t < \theta \Rightarrow \text{Anomaly at time } t \tag{2}$$

This forecasting-as-detection paradigm allows a single architecture to serve both HI estimation and online anomaly detection. Crucially, the forecasting Mean Squared Error (MSE) acts as a dynamic indicator of system health. By applying Extreme Value Theory (EVT) through the Peak-Over-Threshold (POT) method, we provide a mathematically rigorous way to define thresholds that adapt to varying industrial noise levels, a feature missing in traditional 3-Sigma or fixed-percentile methods.

III. RELATED WORK

Recent advancements in Deep Learning (DL) have introduced end-to-end architectures that bypass manual feature engineering for bearing PHM. We organize the related literature along four architectural axes.

A. RECURRENT ARCHITECTURES

Long Short-Term Memory (LSTM) networks have been widely utilized for bearing RUL prediction due to their ability to capture temporal dependencies in sequential vibration data [6]. However, LSTMs often suffer from vanishing gradients when processing high-frequency signals over extended horizons, and their sequential computation is inherently difficult to parallelize. Lai et al. [15] recently proposed a hybrid architecture combining a multi-channel CNN front-end with a Mamba-enhanced adaptive self-attention LSTM, demonstrating improved fault classification accuracy on the Paderborn and CWRU datasets. While this work shares our interest in combining CNN and Mamba components, it focuses on *fault diagnosis* (classification of fault type) rather than *prognostics* (continuous HI forecasting and anomaly detection), and does not employ the Channel-Independent paradigm.

B. CONVOLUTIONAL AND TEMPORAL NETWORKS

Temporal Convolutional Networks (TCN) [7] and their modern variants, such as ModernTCN [8], have emerged as strong alternatives to recurrent models. By employing dilated causal convolutions and large kernel sizes, TCNs achieve large receptive fields while maintaining fully parallelizable training. ModernTCN, presented at ICLR 2024, further improves upon the standard TCN design through large-kernel depthwise convolutions and channel mixing, providing a strong convolutional baseline for long-sequence tasks.

C. ATTENTION-BASED MODELS

Transformers have set new benchmarks in sequence modeling through the self-attention mechanism [9]. Despite their global context awareness, the quadratic complexity $O(L^2)$ relative to sequence length L poses significant challenges for deployment in resource-constrained edge devices in industrial IoT settings. PatchTST [10] introduced a paradigm shift by segmenting time series into patches and applying Transformer self-attention over those patch tokens in a Channel-Independent fashion, substantially improving efficiency and generalization in long-horizon forecasting.

D. STATE-SPACE MODELS AND MAMBA-BASED APPROACHES

The most recent frontier in sequence modeling is the Mamba architecture [11], a structured Selective State-Space Model (S4-based) that achieves linear scaling $O(L)$ with sequence length while matching or exceeding Transformer-level performance on diverse benchmarks. Its hardware-aware design and selective state update mechanism make it particularly attractive for processing long industrial time series.

Several concurrent works have applied Mamba to bearing prognostics. **FEMamba** [13] proposes a Feature-Enhanced Mamba framework that incorporates degradation-stage-aware global regularization to improve RUL prediction; however, it does not employ a Channel-Independent patch-based embedding or address anomaly detection. **TFG-Mamba** [14] fuses temporal and frequency-domain information through a gated Mamba mechanism to improve computational efficiency in bearing RUL regression. While TFG-Mamba also emphasizes efficiency, it employs FFT-based frequency branch fusion rather than convolutional spatial patching, and does not report AUPRC or inference latency as evaluation metrics.

In the broader time-series forecasting literature, Huang et al. [12] demonstrated that a Channel-Independent architecture combining patch embeddings with a Mamba backbone significantly improves long-horizon multivariate forecasting accuracy. However, this approach is designed for general forecasting tasks and does not address the domain-specific requirements of bearing PHM: exponential HI label construction, calibrated anomaly thresholds, or inference latency evaluation for edge deployment.

Multi-scale spatial-temporal Mamba hybrid models have also been explored for tool RUL prediction. **ST-MambaFormer** [16] combines a heavy ResNet-50 CNN backbone with Mamba for multi-scale feature fusion, targeting cutting tool degradation. While the hybrid motivation is similar to our work, the use of a large pre-trained ResNet backbone, a different degradation domain, and the absence of a parameter budget constraint distinguish it substantially from our approach.

FIGURE 1. 0

Research Gap. Existing Mamba-based prognostics works focus on direct RUL regression and do not jointly address: (i) the Channel-Independent paradigm with lightweight multi-scale convolutional patching; (ii) exponential HI-based anomaly detection evaluated with AUPRC; and (iii) parameter-budget-controlled architectural fairness. Our proposed CI-Mamba directly targets this gap.

IV. PROPOSED METHODOLOGY

A. PROBLEM FORMULATION AND DATA PROCESSING

Let $S \in \mathbb{R}^{C \times L_{total}}$ denote the multivariate vibration signal. Following the sliding window paradigm, we generate input-target pairs (X, Y):

- **Lookback (Input)**: $X = S[:, t : t + L] \in \mathbb{R}^{C \times L}$
- **Horizon (Target)**: $Y = S[:, t+L : t+L+K] \in \mathbb{R}^{C \times K}$

where L and K represent the lookback and horizon lengths, respectively.

To handle non-stationarity, each window X is normalized using Adaptive Normalization (RevIN):

$$\mu_X = \frac{1}{L}\sum_{i=1}^{L} X_i, \quad \sigma_X = \sqrt{\frac{1}{L}\sum_{i=1}^{L}(X_i - \mu_X)^2 + \epsilon} \tag{3}$$

$$\hat{X} = \frac{X - \mu_X}{\sigma_X} \tag{4}$$

The same statistics (μ_X, σ_X) are used to normalize the target Y and denormalize the forecast $\hat{Y}$.

B. HYBRIDMAMBA++ ARCHITECTURE

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The architecture of HybridMamba++ consists of three primary modules: (1) Series Decomposition, (2) Multi-scale Patching, and (3) Selective SSM Encoding.

1) Module 1: Series Decomposition

To decouple the stable degradation trends from volatile transient impulses, we apply a series decomposition layer as defined in our pipeline logic:

$$X_{trend} = \text{AvgPool}(X, \text{kernel} = k) \tag{5}$$

$$X_{seasonal} = X - X_{trend} \tag{6}$$

The trend component captures the gradual increase in vibration energy, while the seasonal component focuses on high-frequency shocks.

2) Module 2: Multi-Scale Dual-Kernel Patching

The seasonal branch employs a multi-scale patch embedding $\{P_1, P_2, \ldots, P_m\}$ to capture diverse frequency components:

$$E_{seasonal} = \text{Concat}(\text{PatchEmbed}_{P_1}(X_{seasonal}), \ldots, \text{PatchEmbed}_{P_m}(\tag{7}$$

3) Module 3: Selective SSM Encoding and Mixing

The final forecast $\hat{Y}$ is a weighted sum of the seasonal forecast $\hat{Y}_s$ and trend forecast $\hat{Y}_t$:

$$\hat{Y} = \sigma(\alpha) \cdot \hat{Y}_s + (1 - \sigma(\alpha)) \cdot \hat{Y}_t \tag{8}$$

where α is a learnable parameter per channel.

C. AUTOMATED ANOMALY DETECTION VIA POT

The anomaly score s for each window is calculated as the Mean Squared Error (MSE) of the prediction:

$$s = \frac{1}{C \cdot K} \sum_{c=1}^{C} \sum_{k=1}^{K} (Y_{c,k} - \hat{Y}_{c,k})^2 \tag{9}$$

To define a robust threshold z_q, we utilize the Peak-Over-Threshold (POT) method. Given a high quantile t (e.g., P98), we fit a Generalized Pareto Distribution (GPD) to the excesses $E = \{s_i - t \mid s_i > t\}$. The final threshold z_q for a target probability q is:

$$z_q = t + \frac{\sigma}{\gamma} \left(\left(\frac{n}{N_t} q \right)^{-\gamma} - 1 \right) \tag{10}$$

This ensures a near-zero False Alarm Rate while maintaining high sensitivity.

D. EXPERIMENTAL SETUP

1) Dataset and Preprocessing

Experiments are conducted on the PRONOSTIA (FEMTO-ST) dataset. We follow a 10/50/40 data partitioning strategy: the first 10% (break-in) is discarded, the next 50% (healthy state) is used for training and threshold calibration, and the remaining 40% (degradation state) is used for testing. Signals are normalized using Adaptive Normalization (RevIN) to handle distribution shifts.

2) Baselines and Parameter Budget

To ensure architectural fairness, all models are constrained to a parameter budget of 200k–250k parameters. We compare HybridMamba++ against LSTM, TCN, ModernTCN, and a Patch-based Transformer.

V. EXPERIMENTAL RESULTS AND DISCUSSION

A. QUANTITATIVE PERFORMANCE COMPARISON

The performance of HybridMamba++ and the baselines is summarized in Table 1. HybridMamba++ achieves the highest F1-Score (0.9859) and AUPRC (0.9944), demonstrating its superior capability in distinguishing incipient faults from normal operation.

B. IMPACT OF THRESHOLDING STRATEGIES

The effectiveness of the anomaly detection pipeline is heavily dependent on the thresholding strategy. We compare the proposed POT method against standard statistical baselines, as shown in Table 2. The POT method (q=1e-3) achieves the best balance, providing the highest F1-Score (0.9859) and the lowest False Alarm Rate (2.03%), which is critical for reducing unnecessary maintenance costs in industrial settings.

TABLE 1. Performance comparison on the FEMTO-ST Dataset (Bearing B02).

Method	F1-Score	AUPRC	FAR (%)	Latency (ms)
LSTM	0.9214	0.9423	5.42	0.1542
TCN	0.9458	0.9610	4.18	0.0985
ModernTCN	0.9672	0.9785	3.25	0.0812
Transformer	0.9715	0.9821	2.84	0.1245
HybridMamba++	**0.9859**	**0.9944**	**2.03**	**0.0673**

FIGURE 2. 0

TABLE 2. Comparison of Thresholding Strategies (HybridMamba++ on B02).

Method	F1-Score	FAR (%)	Threshold
3-Sigma	0.9822	2.58	1.4104
Robust (MAD)	0.9851	2.15	1.6010
Percentile (99.7)	0.9852	2.13	1.6171
POT (q=1e-3)	**0.9859**	**2.03**	**1.6928**

C. CORRELATION WITH PHYSICAL DEGRADATION TRENDS

A key advantage of HybridMamba++ is its alignment with the physical degradation process. As shown in our analysis, the Anomaly Score (MSE-based) exhibits a strong correlation with traditional physical indicators such as Root Mean Square (RMS) and Kurtosis. When incipient degradation occurs (at approximately the 335th file in the FEMTO-ST test sequence), the Anomaly Score reacts instantaneously with a sharp jump, preceding total failure. This demonstrates that the model captures meaningful structural changes in vibration signatures rather than mere stochastic noise.

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D. INFERENCE EFFICIENCY FOR EDGE DEPLOYMENT

HybridMamba++ exhibits the lowest inference latency (0.0673 ms per sample), which is 45% faster than the Transformer baseline. This efficiency stems from the linear complexity of the selective SSM mechanism and the lightweight convolutional patch embedding.

VI. CONCLUSION

This paper introduces HybridMamba++, a series-decomposed state-space model for high-precision bearing anomaly detection. By explicitly decoupling trend and seasonal components and employing multi-scale dual-kernel patching, the proposed model effectively captures the complex degradation dynamics of industrial bearings. Experimental results on the FEMTO-ST benchmark confirm that HybridMamba++ achieves state-of-the-art performance in terms of F1-Score, AUPRC, and inference latency under a strict parameter budget. Future work will explore self-supervised pre-training and domain adaptation for varying operating conditions.

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